

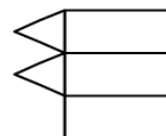
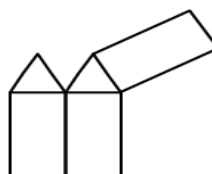
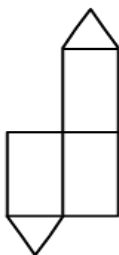
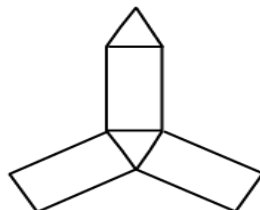


Visualising 3D objects from nets and nets from 3D objects

Teacher resource

Question 1

Which of these can fold to form a triangular prism?



Answer: A

Skill: Students recognise and visualise nets of a prism.

Background information

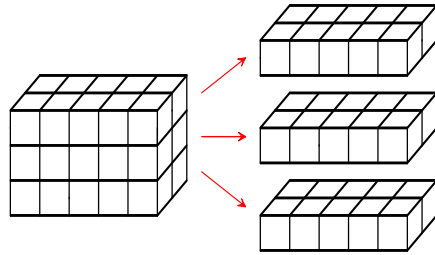
Students should understand that using a net is only one way of producing a three dimensional object. They may need many opportunities to develop the high level visualisation skills that are required to predict the outcome of folding a net. The arrangement of faces in a flat two-dimensional net does not necessarily indicate the position and orientation of faces in the folded net. Students may need multiple opportunities to engage in constructing and folding nets, cutting three dimensional objects to produce nets as well as predicting and testing the effect of rearranging faces of a net.

Additional questions

1. Diagram B could be changed to a net of a triangular prism. Describe how this could be achieved with the minimum number of cuts and transformations.
2. Describe the minimum number of cuts and transformations required to change the other two diagrams to nets of triangular prisms.

Question 2

A large prism is cut into three identical small prisms.



The ratio of the surface area of the large prism to the total surface area of the three small prisms is closest to:

- ☐ 1:2
- ☐ 1:3
- ☐ 2:1
- ☐ 3:1

Answer: A

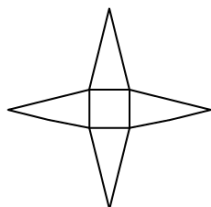
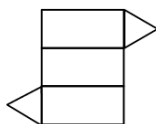
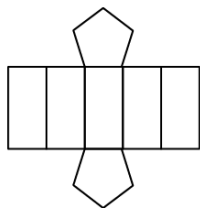
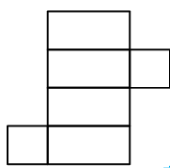
Skill: Students use visualisation skills to support calculation of surface area.

Additional question

A cube is cut into two identical rectangular prisms. What is the ratio of the surface area of one of the prisms to the surface area of the cube?

Question 3

Which of these can fold to form a shape with six vertices and one pair of parallel sides?



Answer: C (a triangular prism)

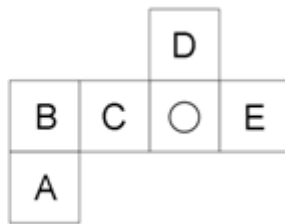
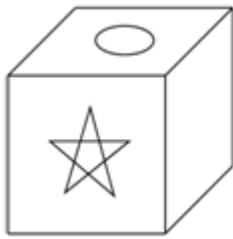
Skill: Students interpret geometric language and visualise 3D objects.

Additional questions

1. Name the 3D objects created by folding each of the nets.
2. Describe the other 3D objects in terms of numbers of faces, number of vertices and pairs of parallel faces.

Question 4

A cube and its net are shown below.



On which face or faces could the star appear given its position on the cube?

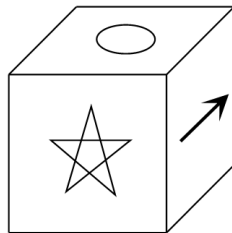
A	B	C	D	E
☐	☐	☐	☐	☐

Answer: A, C, D and E

Skill: Students interpret geometric language to construct 3D objects.

Additional questions

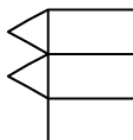
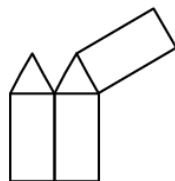
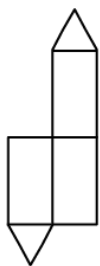
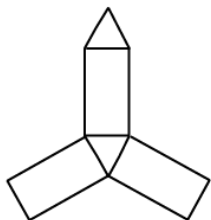
1. For each possible correct answer, draw the star on the net in its correct orientation, for the cube to appear as pictured.
2. Draw three different possible nets for the cube below showing all symbols in both their correct position and orientation.



Student activity: Visualising 3D objects from nets and nets from 3D objects

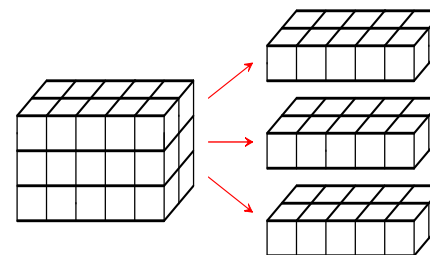
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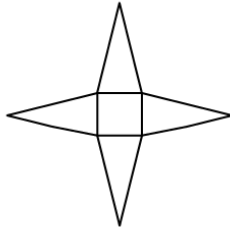
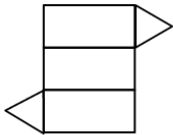
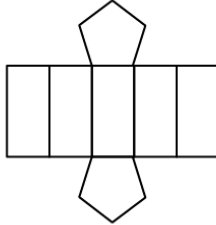
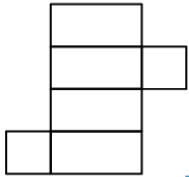
☐ 1:3

☐ 2:1

☐ 3:1

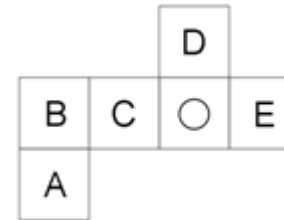
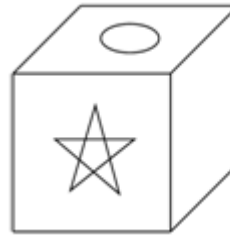
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☐	☐	☐	☐	☐